Science

Class 8 (English Medium)

Solved Exercises — Sindh Textbook Board

12 lessons — 226 questions

انساني عضون وارو سرشتو

Question: 1. (i) The upper part of the bone is called ___.

Answer: The cranium (skull). The brain stays protected inside the cranium.

Question: 1. (ii) The upper part of the brain is called ___.

Answer: The fore-brain (cerebrum). It is the largest part of the brain.

Question: 1. (iii) Cerebellum means ___.

Answer: Little brain. The cerebellum is a part of the hind-brain which controls the balance and movement of the body.

Question: 1. (iv) The hind part of the brain consists of the pons (Pons) and ___.

Answer: The medulla oblongata and the cerebellum. The hind-brain has three parts: the medulla oblongata, the pons and the cerebellum.

Question: 1. (v) The large part of the brain is ___.

Answer: The fore-brain (cerebrum). It consists of two hemispheres.

Question: 1. (vi) The extension of the brain up to the backbone is called ___.

Answer: The spinal cord. It extends from the brain downwards through the backbone.

Question: 1. (vii) The waste thing obtained during the breathing process of the cell is ___.

Answer: Carbon dioxide. It is the waste material produced during the breathing process which is removed from the body.

Question: 1. (viii) The main organ of the excretory process is ___.

Answer: The kidney. The kidneys clean the blood and remove waste materials in the form of urine.

Question: 1. (ix) The artificial kidney machine is called ___.

Answer: The dialysis machine. When a kidney fails, this machine does the work of removing waste materials from the blood.

Question: 1. (x) A completely failed kidney can be managed through ___.

Answer: Kidney transplant. The damaged kidney is removed and a healthy kidney is fitted in its place.

2. (i) The nervous system is a net or network of ___.

✓ (a) Neurons

(b) Spinal cord

(c) Nerves

(d) Lobe

2. (ii) The fore-brain is called ___.

(a) Cerebellum

✓ (b) Cerebrum

(c) Medulla oblongata

(d) Pons

2. (iii) The central nervous system is made of ___.

(a) Cerebrum, thalamus and hypothalamus

(b) Cerebrum, pons and medulla oblongata

✓ (c) Brain, spinal cord and nerves

(d) Cerebellum, brain and nerves

2. (iv) Which is the main organ of the urine excretion process?

✓ (a) Kidney

(b) Liver

(c) Skin

(d) Lung

2. (v) Changing a damaged kidney into a healthy kidney is called ___.

(a) Transformation

✓ (b) Transplantation

(c) Transpiration

(d) Translocation

Question: 3. (i) What work does the nervous system do? How does this system carry messages between the brain and the body?

Answer: The nervous system controls the work of all the systems and organs of the body and creates coordination among them. The sensory nerves carry messages from the different parts of the body to the brain and the spinal cord, the brain decides on them, and the motor nerves carry the brain's orders to the organs.

Question: 3. (ii) The nervous system has control over the actions of our body. Give a list of some actions.

Answer: The nervous system controls these actions: thinking, remembering, seeing, hearing, speaking, walking, breathing, the beating of the heart, the digestion of food and the control of temperature.

Question: 3. (iii) The nervous system has important parts. What work does each one do?

Answer: The nervous system has three important parts: (1) The brain — thinking, understanding, deciding and controlling all actions. (2) The spinal cord — the path for messages between the brain and the body and the centre of automatic actions. (3) The nerves — carrying messages; the sensory nerves carry messages to the brain and the motor nerves carry orders to the organs.

Question: 3. (iv) Give the names of the parts of the brain which control breathing, thinking, emotions and the temperature of the body.

Answer: Thinking and emotions — the fore-brain (cerebrum); breathing and the beating of the heart — the medulla oblongata (hind-brain); control of the body's temperature, hunger and thirst — the hypothalamus.

Question: 3. (v) Explain the following: (a) Frontal lobe (b) Parietal lobe (c)

Temporal lobe (d) Occipital lobe.

Answer: (a) Frontal lobe — producing movement, intelligence, skill, speaking, reasoning, deciding and personality. (b) Parietal lobe — translating the stages of the senses, language, mathematics and help in reading and writing. (c) Temporal lobe — distinguishing hearing- and memory-related information and storing it as memory. (d) Occipital lobe — the work of seeing and recognising the things seen.

Question: 3. (vi) Show the difference between voluntary and involuntary actions.

Answer: Voluntary action — an action done by a person's own will, like walking, eating and speaking; its centre is the fore-brain. Involuntary action — an action that happens automatically without a person's will, like the beating of the heart, breathing and the digestion of food; its centre is the hind-brain.

Question: 3. (vii) Give five examples of reflex action related to daily life.

Answer: (1) The hand pulling back at once when it touches a hot thing. (2) The eye blinking when something falls into it. (3) Sneezing when there is irritation or dust in the nose. (4) Coughing when food gets stuck in the throat. (5) The leg moving forward automatically when there is a light tap below the knee.

Question: 3. (viii) Why does the body need the excretory system?

Answer: The body needs it because during the processes of eating and breathing, waste materials (urea, carbon dioxide, extra water and salts) are produced in the body. If these do not leave the body, they become poisonous and harm the body, so it is necessary to throw them out through the excretory system.

Question: 3. (ix) Explain the importance of the kidney in the excretory system.

Answer: The kidney is the main organ of the excretory system. It cleans the blood, separates waste materials like urea, extra water and salts from the blood and makes urine, and keeps the quantity of water in the body in balance. Without the kidney, waste materials would collect in the blood and make the body poisonous.

Question: 3. (x) Draw a diagram of the human excretory system and label it.

Answer: Draw a diagram showing the two kidneys, the arteries and veins of the kidneys, the ureters, the bladder and the urethra. Put a label with the name on each part.

Question: Project (i) Compare the human brain with a computer and discuss it with the students in the class.

Answer: Like a computer, the brain receives information, stores it and acts on it. The difference is that the brain can itself think and decide, whereas a computer only acts on the instructions given to it. Discuss this comparison in the class.

Question: Project (ii) Visit your nearby hospital and talk to a doctor about the common diseases of the kidney.

Answer: Go to the hospital and ask the doctor about the common diseases of the kidney (like kidney stones, infection, and kidney failure caused by diabetes and high blood pressure); obtain information about their causes, accidents, treatment and death rate.

Question: Project (iii) Understand the dialysis process of a kidney patient and find out from the doctor how many patients wait for a kidney transplant every year.

Answer: Watch and understand the dialysis process of a patient — how the dialysis machine removes waste materials from the blood. Ask the doctor how many patients remain waiting for a kidney transplant every year.

Question: Think and ponder 1. Why is the brain the ruler of the human body?

Answer: The brain is the ruler because it controls the work of all the organs and systems of the body, receives messages from the senses, decides on them and gives orders to the organs.

Question: Think and ponder 2. Why does the brain work all the time, and why does it not stop its work during sleep?

Answer: The brain works all the time because even during sleep the necessary actions of the body continue, like breathing, the beating of the heart and the control of the body's temperature. If the brain stopped working, these actions would stop too.

Question: Think and ponder 3. What is the work of the spinal cord in the nervous system?

Answer: The spinal cord is the path for messages between the brain and the body and the centre of automatic actions (reflex actions).

Question: Think and ponder 4. When the doctor examines the symptoms of kidney-related problems, why does he often advise the patient to get the liver function checked?

Answer: Because the liver also plays an important role in separating waste materials from the blood and cleaning the body, and the work of the kidney and the liver is connected with each other; a fault in the liver can produce symptoms like those of kidney problems.

Question: Think and ponder 5. During exercise most people sweat a lot. One reason is that sweating helps keep the body cool. Give another reason.

Answer: Another reason is that through sweat, extra water and salts and other waste materials also leave the body — that is, the skin also helps in the excretory process.

Question: Think and ponder 6. Many people are born with one kidney, yet they are normal. What necessary precautions should they take?

Answer: They should protect their one kidney: drink plenty of water, avoid excessive use of salt and medicines, keep blood pressure and sugar under control, and get checked by a doctor from time to time.

Question: Model making: Make a model of the brain (joint work).

Answer: Make a model of the brain with dough (or modelling clay) of five different colours: use a separate colour for each lobe (frontal, parietal, temporal, occipital, cerebellum) and give a separate colour to the brain stem. Put a label with the name on each part and explain it in the class.

جاندارن ورثو

Question: 1. (i) Explain the difference between mitosis and cell division.

Answer: Cell division is the general process in which both the nucleus and the cytoplasm divide. Mitosis is the division of the nucleus in which two identical nuclei are formed from one nucleus; after that, in cytokinesis the cytoplasm divides and two new cells are formed.

Question: 1. (ii) What is the meaning of meiosis?

Answer: Meiosis means reduction division. In it the number of chromosomes becomes half, and four cells are formed from one cell.

Question: 1. (iii) What is the purpose of interphase?

Answer: In interphase the cell prepares for division: the DNA is copied (replication), the cell grows, and the materials needed for division are prepared.

Question: 1. (iv) What is the meaning of haploid and diploid?

Answer: Haploid (n) means one set of chromosomes, as in the sex cells (gametes). Diploid (2n) means two sets of chromosomes, as in the ordinary body cells.

Question: 1. (v) Define the following terms: (a) Homologous chromosomes (b) Gene (c) Cytokinesis.

Answer: (a) Homologous chromosomes — the pair of chromosomes in which one chromosome comes from the father and one from the mother, and both are alike in shape, size and genes. (b) Gene — the part of the DNA which carries the information of one hereditary characteristic. (c) Cytokinesis — the division of the cytoplasm in cell division, by which two separate cells are formed.

Question: 1. (vi) Tell the two ways of cell division and explain their

purpose.

Answer: There are two ways: (1) Mitosis — the division of body cells; its purpose is the growth and repair of the body. (2) Meiosis — the division of sex cells; its purpose is to make sex cells (gametes) and to keep the number of chromosomes maintained.

Question: 1. (vii) Differentiate between mitosis and meiosis.

Answer: (1) Mitosis takes place in body cells, meiosis in sex cells. (2) In mitosis there is one division and two cells are formed; in meiosis there are two divisions (I and II) and four cells are formed. (3) In mitosis the number of chromosomes stays the same; in meiosis it becomes half. (4) The cells formed by mitosis are like the parent cell; the cells formed by meiosis are different.

Question: 1. (viii) Define heredity and give examples of some hereditary characteristics.

Answer: Heredity is the way by which the characteristics of the parents are transferred to the offspring. Examples: the colour of the eyes, the colour and texture of the hair, height, the ability to roll the tongue, and whether the ear lobe is attached or free.

Question: 1. (ix) Define DNA and draw its diagram.

Answer: DNA (deoxyribonucleic acid) is the material which carries hereditary information. It is present in the chromosomes in the nucleus, is like a two-stranded ladder (double helix), and is made of nucleotides; each nucleotide contains sugar, phosphate and a nitrogen base (adenine, thymine, guanine or cytosine). Draw a diagram showing the two twisted strands and the ladder-like bases.

Question: 2. (i) Characteristics like the ability to roll the tongue and the ear lobe are received from the parents in ___.

Answer: Through the genes in inheritance. These characteristics are

transferred from parents to offspring through the information present in the genes.

Question: 2. (ii) Chromosomes contain ___.

Answer: DNA. Chromosomes are made of DNA and protein.

Question: 2. (iii) There are two ways of cell division: 1. ___ 2. ___.

Answer: 1. Mitosis 2. Meiosis.

Question: 2. (iv) Prophase means ___.

Answer: The first stage. Prophase is the first stage of mitosis, in which the chromosomes appear.

Question: 2. (v) The transfer of characteristics or traits from parents to offspring is called ___.

Answer: Heredity.

Question: 3. Project: Make a Mitosis Flip Book.

Answer: Draw the stages of mitosis in order on many cards: interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Write the name of the stage on each card. Join the cards in order to make a book; when the pages of the book are flipped quickly, the division of the cell will appear to move.

علم فن حیات

Question: 1. (i) Define biotechnology.

Answer: Biotechnology is the scientific knowledge in which living organisms or their materials are used with technology in an ethical way to produce useful things or solutions to problems.

Question: 1. (ii) Give some examples of products prepared in food and agriculture with the use of biotechnology.

Answer: Examples: genetically changed crops (maize, wheat, rice, cotton, soybean), crops with more nutrition (like rice with vitamin A), bananas prepared through tissue culture, cheese, yoghurt and bread made with yeast.

Question: 1. (iii) Explain the role of bacteria in biotechnology.

Answer: Bacteria are used in genetic engineering: a foreign gene is inserted into their plasmid, and the bacteria produce human insulin, proteins and other things from that gene in large quantity.

Question: 1. (iv) Note down some uses of biotechnology in daily life.

Answer: Daily uses: insulin for diabetes, vaccines for protection from diseases (BCG, measles, polio, hepatitis), growth hormone for dwarfism, the preparation of cheese and yoghurt, genetically changed crops, and bananas through tissue culture.

2. (i) In which microbe is a gene inserted to prepare human insulin on a commercial scale?

(a) Virus

✓ (b) Bacteria

(c) Algae

(d) Fungus

2. (ii) The product of the human growth hormone is used for the treatment of ___.

(a) Night blindness

✓ (b) Dwarfism

(c) Measles

(d) Diabetes

2. (iii) The stage of the copying of DNA is called ___.

(a) Lengthening

(b) Regeneration

✓ (c) Replication

(d) Reproduction

2. (iv) To obtain which vitamin in the rice plant is it genetically changed with the gene of that vitamin?

(a) Vitamin B

(b) Vitamin K

✓ (c) Vitamin A

(d) Vitamin C

2. (v) Which of the following diseases can be treated through the use of a vaccine?

(a) Diabetes

✓ (b) Measles

(c) AIDS

(d) Cancer

2. (vi) In DNA, information is kept in the form of coded writing. It has ___ organic bases.

✓ (a) 4

(b) 5

(c) 6

(d) 2

2. (vii) The small pieces of DNA are called ___.

✓ (a) Plasmid

(b) Replica

(c) Template

(d) Clone

2. (viii) Which scientists proposed the model of DNA in 1953?

(a) Haber and Bosch

(b) Schleiden and Schwann

✓ (c) Watson and Crick

(d) Darwin and Crick

2. (ix) Which of the following weakens the disease-spreading organisms (pathogens)?

(a) Antibody

(b) Plasmid

✓ (c) Vaccine

(d) Antibiotics

2. (x) Which of the following is not an organic base of DNA?

(a) Guanine

(b) Thymine

(c) Adenine

✓ (d) Uracil

Question: 3. Match column (A) with (B): DNA — double-helix structure; replication — copies; separation of insulin — human pancreas; ribose sugar in cytosine — structure of DNA; the way auxin leaves the cell — to increase banana yield. [note: the wording of the first pair is not clear in the book; this pair is made from the remaining boxes.]

Answer: DNA — double-helix structure (the model of DNA is like a ladder of

two twisted strands). Replication — copies (copies of DNA are formed by replication). Separation of insulin — human pancreas (insulin comes out of the human pancreas). Ribose sugar in cytosine — structure of DNA (the structure of DNA includes sugar and organic bases). The way auxin leaves the cell — to increase banana yield [note: this pair is not clear in the book].

آلودگي (گدلاڻ) ان جا ماحول تي اثرات

Question: 1. (i) What are the main air pollutants? Tell their causes and effects on the human body.

Answer: The main air pollutants: sulphur dioxide, carbon monoxide, nitrogen dioxide and chlorofluorocarbon. Causes: smoke from factories and vehicles, the burning of fuel, and the use of refrigerators. Effects: breathing diseases, cough, cold, headache, heart diseases, damage to the nervous system and spoiling of the lungs.

Question: 1. (ii) Define pollution.

Answer: Pollution is such an unnatural change in the physical, chemical or biological characteristics of air, water and land which harms the environment and the life of living things.

Question: 1. (iii) What is the role of the ozone layer in the survival of the environment?

Answer: The ozone layer protects the earth from the harmful ultraviolet rays of the sun. If this layer were not there, these rays would produce skin cancer and other diseases in living things.

Question: 1. (iv) What is the green house effect?

Answer: The green house effect is the process in which gases like carbon dioxide, methane and chlorofluorocarbon hold the sun's heat on the earth, by which the earth becomes warm and the global temperature rises.

Question: 1. (v) Tell different ways to reduce pollution.

Answer: Ways: (1) Planting more trees. (2) Reducing the use of fuel. (3) Fitting filters on the chimneys of factories. (4) Reusing and recycling things. (5) Reducing the use of chemical fertilisers and insect-killing medicines. (6) Not wasting water. (7) Creating awareness among people

and acting on the law.

2. (i) The extremely delicate and dangerous type of pollution is ___.

(a) Water pollution

✓ (b) Air pollution

(c) Noise pollution

(d) Land pollution

2. (ii) Tell the special cause of the hole in the ozone layer.

(a) Carbon dioxide

(b) Sulphur dioxide

✓ (c) Chlorofluorocarbon

(d) Oxygen

2. (iii) Which gas comes out because of the burning of fuel?

(a) Carbon monoxide

(b) Chlorofluorocarbon

✓ (c) Carbon dioxide

(d) Sulphur dioxide

2. (iv) The layer which protects the earth is called ___.

✓ (a) Ozone

(b) Layer of oxygen

(c) Layer of CFC

(d) Green house

2. (v) Which of the following gases is especially responsible for the green house effect in the environment?

(a) Nitrogen dioxide

(b) Methane

(c) Sulphur dioxide

✓ (d) Carbon dioxide

2. (vi) What thing is included in the destruction of historical buildings?

(a) Flood

(b) Tsunami or sea wave

✓ (c) Acid rain

(d) Green house

2. (vii) The process by which we can save the natural resources of the earth:

(a) Wasting water

(b) Engine of gas in the air

✓ (c) Reusing and recycling things

(d) Cutting of forests

2. (viii) ___ is included among the animals of rare species.

(a) Deer

✓ (b) Panda

(c) Donkey

(d) Cat

2. (ix) The poisonous gas which comes out with a biting smell in the smoke of motor vehicles:

(a) Oxygen

✓ (b) Sulphur dioxide

(c) Silver nitrate

(d) Ammonia

2. (x) The ozone layer is present in the atmosphere at a height of about ___ above the land surface.

(a) 5 to 10 kilometres

(b) 16 to 20 kilometres

✓ (c) 12 to 50 kilometres

(d) 16 to 50 kilometres

Question: 3. Match column (A) with (B): saving the earth — save the earth;

cutting of trees for ever — cutting of forests; excessive use of chemical fertiliser — pollution; drowsiness and unclear speech — effect of chlorofluorocarbon; environmental problem — ozone layer. [note: the last pair is not clear in the book; this pair is made from the remaining boxes.]

Answer: Saving the earth — save the earth. Cutting of trees for ever — cutting of forests (the cutting of trees is called deforestation). Excessive use of chemical fertiliser — pollution (excess chemical fertiliser produces pollution in land and water). Drowsiness and unclear speech — effect of chlorofluorocarbon (symptoms like drowsiness and unclear speech occur from the effect of CFC). Environmental problem — ozone layer [note: this pair is not clear in the book].

كيميائي عمل

Question: 1. Define a chemical reaction and write examples from daily life.

Answer: A chemical reaction is the process in which one or more things (reactants) change and make new things (products) with new characteristics, and the original thing is not got back. Examples: the digestion of food, milk turning sour, rusting of iron, the burning of a matchstick, and photosynthesis.

Question: 2. Write the four conditions that show a chemical reaction.

Answer: (1) Formation of a new thing with new characteristics. (2) Absorption or release of energy (heat or light). (3) Change in colour. (4) Production of gas or bubbles, or the coming of a smell.

Question: 3. (i) A lock rusting — physical (P) or chemical (C) change?

Answer: (C) Chemical change. Rust is a new material which is formed by the combination of iron and oxygen.

Question: 3. (ii) Melting of ice — physical (P) or chemical (C) change?

Answer: (P) Physical change. Ice melts and becomes water, but the chemical characteristic of water stays the same.

Question: 3. (iii) Milk turning sour — physical (P) or chemical (C) change?

Answer: (C) Chemical change. New materials are formed in the milk and the milk does not come back to its original state.

Question: 3. (iv) Digestion of food — physical (P) or chemical (C) change?

Answer: (C) Chemical change. In digestion, food breaks and makes new materials which are absorbed in the body.

Question: 3. (v) Meat kept in sunlight rotting — physical (P) or chemical (C) change?

Answer: (C) Chemical change. When meat rots, new smelly materials are

formed in it and the meat does not come back to its original state.

Question: 3. (vi) Making a design on a piece of copper — physical (P) or chemical (C) change?

Answer: (P) Physical change. By making a design, the shape of the copper changes but its chemical characteristic stays the same.

Question: 3. (vii) Mixing sugar in water — physical (P) or chemical (C) change?

Answer: (P) Physical change. Sugar dissolves in water but no new material is formed.

Question: 3. (viii) Grinding a stone into powder — physical (P) or chemical (C) change?

Answer: (P) Physical change. By grinding a stone into powder, only its size and shape change.

Question: 3. (ix) Burning of a matchstick — physical (P) or chemical (C) change?

Answer: (C) Chemical change. By burning, new materials (ash, gases) are formed and the matchstick is not got back.

Question: 4. (i) Balance the following chemical equation: $S8 + O2 \rightarrow SO2$

Answer: $S8 + 8O2 \rightarrow 8SO2$

Question: 4. (ii) Balance the following chemical equation: $C2H6 + O2 \rightarrow CO2 + H2O$

Answer: $2C2H6 + 7O2 \rightarrow 4CO2 + 6H2O$

Question: 4. (iii) Balance the following chemical equation: $P4 + O2 \rightarrow P2O5$

Answer: $P4 + 5O2 \rightarrow 2P2O5$

Question: 4. (iv) Balance the following chemical equation: $Ag + S8 \rightarrow Ag2S$

Answer: $16Ag + S8 \rightarrow 8Ag2S$

Question: 4. (v) Balance the following chemical equation: $Al + Br2 \rightarrow AlBr3$

Answer: $2Al + 3Br2 \rightarrow 2AlBr3$

Question: 4. (vi) Balance the following chemical equation: Cr + O2 → Cr2O3

Answer: 4Cr + 3O2 → 2Cr2O3

Question: 4. (vii) Balance the following chemical equation: NaClO3 → NaCl + O2

Answer: 2NaClO3 → 2NaCl + 3O2

Question: 4. (viii) Balance the following chemical equation: AlBr3 + Cl2 → AlCl3 + Br2

Answer: 2AlBr3 + 3Cl2 → 2AlCl3 + 3Br2

5. (i) A synthesis reaction is an example of what?

(a) Carbon dioxide breaking into carbon and oxygen

(b) Zn + 2HCl → ZnCl2 + H2

✓ (c) Hydrogen and oxygen joining to make water

5. (ii) What is the best sign of a chemical reaction taking place?

(a) Blue colour changes water to blue colour

(b) A white thing dissolving in water

✓ (c) Bubbles forming when a metal is put in acidic water

5. (iii) What are obtained in this process? CH4 + 2O2 → CO2 + 2H2O

✓ (a) Carbon dioxide and water

(b) Carbon monoxide and hydroxide

(c) Salt and vinegar

5. (iv) In the following, a chemical reaction is an example of which?

(a) Formation of common salt as sea water evaporates

(b) Paper becoming soft when it gets wet

✓ (c) Leaves changing colour in autumn

Question: 6. Explain the difference between exothermic and endothermic processes.

Answer: Exothermic process — in which heat comes out, like the burning

of fuel and breathing: 2H2 + O2 → 2H2O + heat. Endothermic process — in which heat is absorbed, like the decomposition of mercury oxide: 2HgO + heat → 2Hg + O2, and photosynthesis.

Question: 7. Give the difference between addition reaction and decomposition reaction with examples.

Answer: Addition reaction — in which two or more things join and make one compound, like 2Na + Cl2 → 2NaCl and C + O2 → CO2. Decomposition reaction — in which one compound breaks and makes two or more things, like 2H2O → 2H2 + O2 and 2KClO3 → 2KCl + 3O2.

Question: 8. Explain the importance of exothermic processes.

Answer: Exothermic processes give us heat and energy: the burning of fuel gives heat for cooking food, warming houses and running engines; from breathing (respiration) the body gets energy with which we can work.

Question: 9. (a) Write the following chemical process as a balanced equation: carbon dioxide (CO2) and water (H2O) — photosynthesis.

Answer: 6CO2 + 6H2O + sunlight → C6H12O6 + 6O2

Question: 9. (b) Write the following chemical process as a balanced equation: carbon monoxide (CO) and oxygen (O2).

Answer: 2CO + O2 → 2CO2

Question: 9. (c) Write the following chemical process as a balanced equation: zinc (Zn) and hydrochloric acid (HCl).

Answer: Zn + 2HCl → ZnCl2 + H2

Question: 9. (d) Write the following chemical process as a balanced equation: hydrochloric acid (HCl) and sodium hydroxide (NaOH).

Answer: HCl + NaOH → NaCl + H2O

Question: 9. (e) Write the following chemical process as a balanced equation: the decomposition of potassium chlorate (KClO3).

Answer: 2KClO3 → 2KCl + 3O2

تیزاب، اساس لوٹیات

Question: Activity: Use of indicators for identification — three beakers contain HCl, NaOH and NaCl (not labelled). Test with phenolphthalein, methyl orange and litmus.

Answer: Phenolphthalein: stays colourless in acid (HCl), turns pink in base (NaOH), stays colourless in salt (NaCl). Methyl orange: orange in acid, yellow in base. Litmus: red in acid, blue in base. The beaker in which phenolphthalein turns pink is NaOH, the one in which methyl orange turns orange is HCl, and the remaining neutral salt is NaCl.

Table: Look at the pH of four things — blood 7.3, orange juice 4, ammonia 11, milk 6.5. (i) What describes milk?

(a) Slightly basic

(b) Strongly basic

✓ (c) Slightly acidic

(d) Strongly acidic

(ii) Which list of things is from less to more acidic?

✓ (a) Ammonia, blood, milk, orange juice

(b) Orange juice, milk, blood, ammonia

(c) Ammonia, milk, blood, orange juice

(d) Ammonia, milk, blood, orange juice

(iii) Which thing is more basic?

(a) Blood

(b) Orange juice

✓ (c) Ammonia

(d) Milk

(iv) Which thing's pH is near neutral?

(a) Milk

✓ (b) Blood

(c) Ammonia

(d) Orange juice

(v) Which of the following can be most useful for testing an acidic or basic thing?

(a) Filter paper

✓ (b) Litmus paper

(c) Thermometer

(d) Electric balance

قوت دباؤ

1. (i) Force has ___.

(a) Magnitude

(b) Direction

✓ (c) Magnitude and direction

(d) None

1. (ii) What is the unit of pressure in the international standard?

(a) Joule

(b) Kilogram per square metre

(c) Newton

✓ (d) Pascal

1. (iii) Pressure increases when:

(a) The area of the surface increases

✓ (b) The area of the surface decreases

(c) The magnitude of the force decreases

(d) The area of the surface stays the same

1. (iv) Pressure in water is transferred equally. This rule is called?

(a) Archimedes' principle

✓ (b) Pascal's law

(c) Bernoulli's principle

(d) Le Chatelier's principle

1. (v) Atmospheric pressure can be calculated:

(a) With the height of the water column in a barometer

✓ (b) With the height of the mercury column in a barometer

(c) With the height of the lime water in a barometer

(d) With the height of the oil in a barometer

1. (vi) The quantity of pressure in water increases because of:

(a) Volume

✓ (b) Depth

(c) Base area

(d) Mass

1. (vii) At a height above or below sea level, inner discomfort is felt, because:

(a) The pressure of our body stays the same

✓ (b) The pressure of our body adjusts slowly

(c) The pressure of our body adjusts quickly

(d) Our body is not able to adjust to pressure

1. (viii) Place a rectangular piece of wood in three different ways. The pressure of the piece of wood on the table:

(a) Will be more at place (A)

(b) Will be more at place (B)

✓ (c) Will be more at place (C)

(d) Will stay the same in all three conditions

1. (ix) Which liquid is used in a hydraulic power system?

(a) Water

(b) Oil

✓ (c) Any incompressible liquid

(d) All of these

1. (x) Pressure in a gas is because of ___.

✓ (a) Molecules colliding with the walls of the closed container

(b) Atmosphere (air)

(c) Molecules colliding at the bottom of the closed container

(d) Molecules colliding with each other

1. (xi) This is a mercury barometer instrument which is used to find

atmospheric pressure. The atmospheric pressure is:

(a) 25 cm

(b) 75 cm

(c) 80 cm

(d) 100 cm

Question: 2. A large aquarium is filled with water. The weight of the water in the aquarium is 10000 newtons. The area of the bottom of the aquarium is 1.6 square metres. Find the pressure applied by the water on the bottom of the aquarium.

Answer: Pressure = force ÷ area = 10000 ÷ 1.6 = 6250 pascals. Answer: 6250 Pa.

Question: 3. Suppose that at this time the pressure is 101200 pascals. If the area of the palm of your hand is 0.006 square metres, find the force applied on your hand.

Answer: Force = pressure × area = 101200 × 0.006 = 607.2 ≈ 607 newtons. Answer: 607 newtons.

Question: 4. Look at the diagram given with the copy and answer this question: if you apply a force of 46 newtons on the cylinder and you press it down, then if the area of the piston is 0.5 square metres, what will be the pressure inside the cylinder?

Answer: Pressure = force ÷ area = 46 ÷ 0.5 = 92 newtons per square metre. Answer: 92 N/m².

Question: 5. Define aerosols and explain the working of aerosols.

Answer: Aerosols are very small particles (liquid or solid) floating in air or gas, like the spray coming out of a spray can. Working: in an aerosol can, liquid and gas under pressure are closed. When the valve is pressed, the gas pushes the liquid out of the nozzle, and the liquid comes out in the form of very small particles (spray).

Question: 6. What are other examples of aerosols?

Answer: Examples: hair spray, perfume spray, insect-killing spray, paint spray, the breathing inhaler (medicine for asthma), and the spray that spreads fragrance in a room.

Question: 7. A force of 300 newtons produces a pressure of 4 newtons per square metre. Find the area on which the force is applied.

Answer: Area = force ÷ pressure = 300 ÷ 4 = 75 square metres. Answer: 75 m^2.

Question: 8. State Pascal's law and explain the use of this law with an example.

Answer: Pascal's law: the pressure applied on a closed liquid is transferred equally to every part of the liquid and to the walls of the container.

Example: the braking system of a car and the hydraulic jack — the small force applied on the brake pedal changes through the liquid into a large force which lifts a heavy thing up.

Question: Project: Make a model of a simple barometer.

Answer: Materials: a glass bottle, water, a straight tube (straw) or glass tube, tape and a paper with a scale. Method: fill the bottle with water; fix the tube with tape on the mouth of the bottle so that the lower end of the tube is dipped in the water and the upper end is outside. Fix the paper with the scale in front of the tube. Every day look at the level of the water in the tube and note it: when the pressure of the air increases, the level of the water goes down, and when the pressure decreases, the level goes up.

طبعي مقدارن جي پيمائش

1. (i) A quantity which can be measured is called a ___ quantity.

✓ (a) Physical

(b) Measuring goose

(c) Standard

(d) Dependent on temperature

1. (ii) Which of the following is a base unit?

(a) Newton

✓ (b) Second

(c) Watt

(d) Joule

1. (iii) When a standard is fixed for a quantity, then that standard quantity is called ___.

(a) Price

(b) Rate

(c) Prefix

✓ (d) Unit

1. (iv) What is the correct order of the increasing series for measuring length in increasing size?

(a) Kilometre, centimetre, metre

(b) Metre, centimetre, millimetre

(c) Millimetre, metre, centimetre

✓ (d) Millimetre, centimetre, metre

1. (v) A mass of 4 kilograms is equal to ___.

(a) 4000 milligrams

✓ (b) 4000 grams

(c) 400 grams

(d) 40 grams

1. (vi) A distance of 0.05 kilometres is equal to ___.

(a) 50 centimetres

(b) 5000 metres

(c) 15 metres

✓ (d) 50 metres

1. (vii) 10^{-3} seconds is called ___ .

(a) Microsecond

(b) Nanosecond

(c) Macrosecond

✓ (d) Millisecond

1. (viii) What is the unit of electric current in the international system?

✓ (a) Ampere

(b) Kelvin

(c) Second

(d) Coulomb

1. (ix) The unit of cubic metre is:

✓ (a) Volume

(b) Length

(c) Area

(d) Mass

1. (x) A measuring cylinder is used in the laboratory for?

✓ (a) Measurement of liquid

(b) Measurement of weight

(c) Measurement of coldness

(d) Measurement of volume

Question: 2. What are the parts which together make a physical quantity?

Answer: A physical quantity is made of two parts: (1) the numerical value (example: 3) and (2) the unit (example: metre). Example: in a length of 3 metres, 3 is the numerical value and metre is the unit.

Question: 3. What are the seven base units of physical quantities?

Answer: The seven base units: length — metre (m), mass — kilogram (kg), time — second (s), electric current — ampere (A), temperature — kelvin (K), amount of substance — mole (mol), luminous intensity — candela (cd).

Question: 4. What is a prefix? How can a prefix be changed back into its different international units?

Answer: A prefix is the word or symbol which is placed before a unit and shows a large or small part of the unit, like kilo (1000) and milli (1/1000). Changing: the prefix can be changed by multiplying or dividing by its numerical value, for example: 1 kilometre = 1000 metres, 1 millimetre = 1/1000 metre.

Question: 5. How can you say that the measurement of volume with a graduated cylinder is correct?

Answer: (1) Keep the cylinder on a level surface. (2) Read with the eye level with the surface (meniscus) of the liquid. (3) Read the lower meniscus (lower curve). (4) Use the correct scale and correct mark of the cylinder. If these things are correct, the measurement is correct.

Question: 6. Complete the following table: base physical quantity — name of unit — symbol of unit.

Answer: Length — metre — m; electric current — ampere — A; mass — kilogram — kg; time — second — s; temperature — kelvin — K; amount of substance — mole — mol; luminous intensity — candela — cd.

Question: 7. Five spheres are placed on a measuring strip between two pieces of wood. Calculate what the diameter of one sphere is.

Answer: In the picture, the five spheres spread from 4 cm to 16 cm — that is, the total length of the five spheres = 16 − 4 = 12 cm. The diameter of one sphere = 12 ÷ 5 = 2.4 cm.

گرمي جي توانائي جا وسيلا اثرات

Question: 1. (i) Define thermal expansion. Explain the expansion of metals with the help of an activity.

Answer: Thermal expansion is the process in which a thing expands (becomes bigger) on being heated. Activity: heat a metal wire or ball and it becomes longer or bigger and does not pass through the ring; on cooling, it shrinks back and becomes small.

Question: 1. (ii) What are the effects of the anomalous expansion of water?

Answer: Water, on cooling below 4 degrees, expands instead of shrinking, and expands even more on becoming ice. Effects: ice, being lighter than water, floats on top, so in winter the water under the surface of a pond stays liquid and aquatic life stays alive; in the cold, water freezing bursts pipes.

Question: 1. (iii) Prove with the help of an experiment that gases expand on heating and contract on cooling.

Answer: Experiment: fix a balloon on the mouth of an empty bottle and keep the bottle in hot water — the air inside the bottle becomes hot and expands and inflates the balloon. Keep the bottle in cold water and the air contracts and the balloon sits down.

Question: 1. (iv) Explain the effects of the expansion and contraction of metal things. How can these harmful effects be reduced?

Answer: Effects: metals expand in heat and contract in cold, by which railway tracks can bend, bridges and roads can break, and pipes can burst. Reducing: leaving gaps in railway tracks, fitting expansion joints in bridges, and making crack control joints in roads.

Question: 1. (v) How does a bimetallic strip work in a thermometer?

Answer: A bimetallic strip is made of strips of two different metals which expand by different amounts on heating. On heating, one metal expands more and the strip bends; this bending drives the measurement of temperature or an electric switch (thermostat).

Question: 1. (vi) What problems occur in bridges and railway tracks because of thermal expansion?

Answer: Problems: in heat, railway tracks expand and bend, and there is a danger of a railway accident; the parts of bridges expand and increase pressure, and the bridge can be harmed. For this, gaps and expansion joints are left.

Question: 1. (vii) Why do telephone wires hang down during summer days?

Answer: Because in heat, the metal wires expand and become longer, so they hang down between two poles.

Question: 1. (viii) Why is a gap left in railway tracks?

Answer: Because in the heat of summer, railway tracks expand. If a gap were not left, the tracks would bend and an accident could happen; the gap gives room for expansion.

Question: 1. (ix) What is the strange characteristic of water freezing (becoming ice)?

Answer: The strange characteristic is that water, on freezing, expands instead of shrinking, so ice, being lighter than water, floats on top. Unlike ordinary things, water starts expanding even on cooling below 4 degrees.

2. (i) In which of the following things is expansion and contraction not used?

✓ (a) Electric fan

(b) Electric iron

(c) Railway track

(d) Thermometer

2. (ii) Which liquid is used in a two-walled thermometer?

(a) Water

(b) Oil

✓ (c) Mercury

(d) Vinegar

2. (iii) Which of the following things expands more on the same increase of temperature?

✓ (a) Air

(b) Water

(c) Copper

(d) Glass

2. (iv) A bimetallic strip is used in ___:

(a) Electric fan

(b) Tape recorder

✓ (c) Electric iron

(d) Computer

2. (v) Riveting is used to join ___ firmly.

(a) A soft thing with a hard thing

(b) A soft thing with a soft thing

✓ (c) A hard thing with a hard thing

(d) Plastic with a soft thing

Question: 3. (a) Give the reason: mercury is used in a thermometer.

Answer: Because mercury expands uniformly with heat, does not stick to glass, takes the temperature quickly, and is clearly visible because it is shiny.

Question: 3. (b) Give the reason: a bimetallic strip is used in electric parts.

Answer: Because the rate of expansion of the two metals is different, so on heating the strip bends and the electric circuit is automatically closed or opened (the work of a thermostat).

Question: 3. (c) Give the reason: after heating an iron ball, it does not pass through the ring.

Answer: Because on heating, the iron ball expands and becomes bigger, so it does not pass through the hole of the ring; on cooling, it shrinks and passes through again.

Question: 3. (d) Give the reason: when water freezes in pipes, the pipes burst.

Answer: Because water expands on freezing and the volume of ice becomes more than that of water, so the pressure on the walls of the pipe increases and the pipe bursts.

بلور (لینس)

1. (i) A horizontal line which passes through a lens is called:

(a) Vertical axis

✓ (b) Principal axis

(c) Optical centre of the lens

(d) Focal point

1. (ii) The point through which a ray of light passes without changing its path is:

(a) Centre of curvature

(b) Midpoint of (F) and (O)

(c) Focal point

✓ (d) Optical centre of the lens

1. (iii) To obtain a clear image in a camera:

✓ (a) The lens is moved forward and backward.

(b) The film is moved forward and backward.

(c) Both the lens and the film are moved.

(d) Neither the lens nor the film is moved.

1. (iv) Which of the following parts helps in changing the shape (form) of the lens in the eye?

✓ (a) Ciliary muscles

(b) Cornea

(c) Retina

(d) Iris

1. (v) The distance between the optical centre of the lens and the focal point is called ___:

(a) Principal axis

(b) Vertical axis

(c) Focal length

(d) Optical centre of the lens

Question: 2. (i) When the object is at infinity, the rays are parallel and, passing through a ___ lens, they meet at the focal point.

Answer: Convex lens. The parallel rays coming from infinity pass through a convex lens and meet at the focal point (F).

Question: 2. (ii) An erect image, diminished in size from the original, is formed when the object is placed at any distance in front of a concave lens.

Answer: At any distance. A concave lens always forms an erect, virtual and diminished image, wherever the object is placed.

Question: 2. (iii) A ___ lens is used as a magnifying glass to show things enlarged.

Answer: Convex lens. A convex lens shows things enlarged, so it is used as a magnifying glass.

Question: 2. (iv) The light of distant things focuses in front of the retina of the eye, then a ___ defect is produced, it is called ___.

Answer: A defect of weak eyesight — short sightedness (myopia). In short sightedness, the light of distant things focuses in front of the retina and distant things are not seen clearly.

Question: 2. (v) In long sightedness (hypermetropia), the near point of the eye moves very far away and the light from a near object focuses behind the retina of the eye. [note: the wording of this fill-in question is not clear in the book; the answer is given according to the chapter's material on long sightedness.]

Answer: Near point — far. In long sightedness, the near point of the eye moves far away, so near things are not seen clearly and the light focuses

behind the retina.

Question: 3. Complete the following table: parts of the camera — the same parts of the eye — the same work: (1) Aperture of the camera — pupil/iris of the eye — controlling the quantity of light; (2) Lens — lens of the eye — focusing the light and making the image; (3) Film — retina of the eye — obtaining and sensing the image.

Answer: (1) Aperture of the camera — pupil of the eye — controlling the quantity of light. (2) Lens — lens of the eye — focusing the light and making the image. (3) Film — retina of the eye — obtaining the image.

Question: 4. (i) Explain the process of image formation with the help of a ray diagram of a convex lens.

Answer: Ray diagram in a convex lens: (1) A ray parallel to the principal axis passes through the lens and goes through the focal point (F). (2) A ray passing through the optical centre (O) goes straight. (3) A ray passing through the focal point becomes parallel to the axis after the lens. The image is formed where these rays meet.

Question: 4. (ii) Explain the result of image formation through a concave lens.

Answer: Through a concave lens, the rays spread out and meet on being extended backwards, so the image is always virtual, erect and smaller than the original object, and the image is formed on the same side of the lens on which the object is.

Question: 4. (iii) How does the thickness of a lens affect its focal length?

Answer: The thicker (more curved) the lens is, the more it bends the rays and the shorter its focal length becomes. The thinner the lens is, the longer its focal length becomes.

Question: 4. (iv) Draw a ray diagram for the defect of short sightedness.

Answer: Draw a diagram: the parallel rays coming from a distant object

pass through the lens of the eye and focus in front of the retina of the eye, so a blurred image is formed on the retina. In the treatment diagram, a concave lens spreads the rays so that they focus on the retina.

Question: 4. (v) What type of lens is present in your eye?

Answer: A convex lens is present in the eye, which focuses the light and makes an image on the retina of the eye.

Question: 4. (vi) Write the difference between the defect of short sightedness and long sightedness.

Answer: Short sightedness (myopia): distant things are not seen clearly, light focuses in front of the retina, and spectacles with concave lenses are used for treatment. Long sightedness (hypermetropia): near things are not seen clearly, light focuses behind the retina, and spectacles with convex lenses are used for treatment.

Question: 5. In the given Venn diagram, write the differences and similarities between convex lenses and concave lenses.

Answer: Similarities: both are lenses, both are transparent, both bend light (refraction), and both have an optical centre, a principal axis and a focal point. Difference of the convex lens: thick in the middle and thin at the edges, it converges rays, it forms both real and virtual images, and it is used in a magnifying glass and a camera. Difference of the concave lens: thin in the middle and thick at the edges, it spreads rays, it always forms a virtual, erect and diminished image, and it is used in spectacles for short sightedness and in a door peep hole.

برقي كارج

1. (i) Which part of a bicycle generator rotates to produce electricity?

(a) Qaitio

✓ (b) Magnet

(c) Coil of wire

(d) Galvanometer

1. (ii) The electricity obtained from a cell is called:

✓ (a) Direct current

(b) Indirect current

(c) Alternating current

(d) High current

1. (iii) The electricity which is used in our homes and schools is called what?

(a) Low current

(b) Indirect current

✓ (c) Alternating current

(d) Direct current

1. (iv) Which of the following is not a source of renewable energy?

(a) Wind energy

(b) Geothermal energy

(c) Wave energy

✓ (d) Natural gas

1. (v) The electronic part which can be used as a switch or as a voltage amplifier is:

✓ (a) Transistor

(b) Thermometer

(c) Heat fork flag

(d) Transformer

1. (vi) What is the abbreviation of IC?

(a) Inner Chip

(b) Integrated Chip

✓ (c) Integrated Circuit

(d) Input Circuit

1. (vii) Silicon is an example of ___.

(a) Insulator

(b) Conductor

✓ (c) Semiconductor

(d) Semi-insulator

1. (viii) A barcode is a ___.

(a) Output device

✓ (b) Input device

(c) Amplifier

(d) Insulator

1. (ix) The flow of electrons is called ___.

(a) Electrode

(b) Resistance

(c) Electro light

✓ (d) Electric current

1. (x) When the waste of animals and plants, which is buried in the ground in the absence of air, what is that gas called?

(a) Oxygen

✓ (b) Biogas

(c) Carbon dioxide

(d) Methane

Question: 2. The stages involved in obtaining electricity in a power station are given in disordered form. Write the given boxes in correct order: (a) Water rotates the turbines. (b) Electricity is produced and is included in the national grid. (c) Fuel burns and heats the water tank. (d) Turbines turn the generators. (e) Water changes into steam under pressure.

Answer: Correct order: (c) Fuel burns and heats the water tank → (e) Water changes into steam under pressure → (a) Water (steam) rotates the turbines → (d) Turbines turn the generators → (b) Electricity is produced and is included in the national grid.

Question: 3. Electricity production through wind mills is increasing day by day. Write the answer below: wind energy → ___ → energy of the motion of the rotating blades → ___ → energy obtained from the turbine (output).

Answer: Wind energy → wind pushes and rotates the blades → energy of the motion of the rotating blades → the blades rotate the generator and produce electricity → energy obtained from the turbine (output).

Question: 4. Why are bio-fuels called a cheap source of energy?

Answer: Because bio-fuels are made from the waste and dirt of animals and plants, which are free and found everywhere, and they are renewable sources which can be obtained again and again.

Question: 5. What are the basic components made in electric circuits?

Answer: The basic components: the source of electricity (cell or battery), connecting wires, a switch, a bulb or another load (like a motor), and a resistor. With these, the circuit is completed and electricity flows.

Question: 6. What problems are included in the production of electricity through the following source: (i) Nuclear power generator (ii) Thermal power generator.

Answer: (i) Nuclear power generator: the radioactive waste is dangerous and difficult to keep safe, there is a danger of accident, and setting up the

plant is expensive. (ii) Thermal power generator: the burning of fuel produces pollution in the air and carbon dioxide, the fuel (coal, gas) is finishing, and fuel is expensive.

Question: 7. What are input and output parts? Give some examples.

Answer: Input parts are those which enter information or orders into a computer, like the keyboard, mouse, barcode reader, scanner and microphone. Output parts are those which show the result of the computer outside, like the monitor, printer and speaker.

Question: 8. Tell the difference between alternating current and direct current.

Answer: Alternating current (AC): its direction changes again and again, it is obtained from a generator and the mains electricity of homes, and its voltage can be increased or decreased with a transformer. Direct current (DC): it flows in one direction only, it is obtained from a cell and a battery, and its voltage cannot be changed easily.

Question: 9. What are the advantages of solar electricity?

Answer: Advantages: sunlight is free and found everywhere, solar electricity is renewable and does not finish, no pollution comes from it, the maintenance of solar panels is little, and electricity can be reached even to far-off areas.

خلا جي تلاش

1. (i) A spectroscope is used for:

(a) Discovering the sound waves coming out of a star.

✓ (b) Identifying the chemical elements present in a star.

(c) Bending the light of a star and collecting it at one point.

(d) Marking the location of a star.

1. (ii) A reflecting telescope consists of:

✓ (a) Primary and secondary mirrors.

(b) Only one mirror.

(c) Many lenses.

(d) A prism for the division of light.

1. (iii) Which Moon command module took humans to the Moon?

(a) Apollo 13

(b) Apollo 15

(c) Apollo 16

✓ (d) Apollo 17

1. (iv) Magnetic resonance imaging (MRI) and computed tomography (CT) were obtained as a result of understanding the images of:

(a) Saturn

✓ (b) The Moon

(c) Jupiter

(d) The Sun

1. (v) The Global Positioning System (GPS) consists of how many artificial satellites:

(a) More than 15

(b) More than 20

(c) More than 30

(d) Less than 10

Question: 2. (a) The word telescope is a combination of ___ words.

Answer: Two Greek words. Telescope is made of two Greek words, "tele" (far) and "scope" (to see).

Question: 2. (b) In a reflecting telescope, ___ are used.

Answer: Mirrors are used. In a reflecting telescope, mirrors are used to reflect the light.

Question: 2. (c) The wavelength of ___ colour is the shortest in the strip or spectrum of white light.

Answer: Violet colour. In the spectrum of white light, the wavelength of violet colour is the shortest.

Question: 2. (d) A spacecraft can be ___ or it can also be ___.

Answer: With humans, or without humans. Spacecraft are of both kinds — manned and unmanned.

Question: 2. (e) MRI is the abbreviation of ___.

Answer: Magnetic Resonance Imaging. MRI means Magnetic Resonance Imaging.

Question: 2. (f) Every satellite of the Global Positioning System completes ___ orbits in 24 hours.

Answer: Two orbits. Every GPS satellite completes one full orbit in half a day, so it completes two orbits in 24 hours.

Question: 3. 1. Define the following terms: (a) Telescope (b) Reflecting telescope (c) Refracting telescope (d) Spectroscope (e) Space exploration (f) Space debris.

Answer: (a) Telescope — the instrument which shows far-away things close. (b) Reflecting telescope — the telescope which uses mirrors to reflect light. (c) Refracting telescope — the telescope which uses lenses to

bend light. (d) Spectroscope — the instrument which divides light into a spectrum and identifies the chemical elements present in a star. (e) Space exploration — the process of searching space and using it. (f) Space debris — the rubbish of useless and damaged things in space, like old satellites and pieces of rockets.

Question: 3. 2. How is a reflecting telescope different from a refracting telescope?

Answer: A reflecting telescope uses mirrors to reflect light, whereas a refracting telescope uses lenses to bend light. A reflecting telescope has primary and secondary mirrors, and a refracting telescope has two lenses.

Question: 3. 3. Explain the structure and working of a spectroscope?

Answer: Structure: a spectroscope has three main parts: (1) an opaque barrier with a slit which makes a thin ray of light, (2) a prism which divides the light into its seven colours, (3) a screen or detector which records the light obtained. Working: light passes through the slit and falls on the prism, the prism divides it into a spectrum of seven colours, and from that spectrum the chemical elements present in a star are identified.

Question: 3. 4. Write any five benefits of space exploration in the field of health and treatment?

Answer: (1) Imaging technologies like MRI and CT came from space research. (2) The heart pump for heart patients was made from space technology. (3) The cooling suit is used to keep the body temperature of patients under control. (4) With robot technology, treatment (telemedicine) in far-off areas became possible. (5) From space research, new knowledge about medicines and the human body was obtained.

Question: 3. 5. Which technological instruments are used in space exploration?

Answer: Instruments: telescopes, spectroscopes, spacecraft, artificial

satellites, robots, telecommunication instruments, solar cells, GPS receivers, cameras and other instruments.

Question: 3. 6. Write a short note on the following: (a) The life of an astronaut in space (b) Problems arising because of space exploration (c) The Global Positioning System.

Answer: (a) The life of an astronaut in space: the astronaut wears a special suit, lives in a space station, eats special food and works in weightlessness. (b) Problems arising because of space exploration: space debris (old satellites and pieces of rockets) is increasing, which is a danger for new satellites and spacecraft, and space missions are expensive. (c) The Global Positioning System (GPS): this is a system of artificial satellites which tells the correct location, route and time of any place on the earth; it has 30 or more satellites, and every satellite completes two orbits in 24 hours.

Question: 3. 7. Write the names and work of the different important parts of an artificial satellite / spacecraft?

Answer: (1) Solar panels — producing electricity from sunlight. (2) Cameras and other devices — making scientific measurements and taking pictures. (3) Antennas — sending and receiving signals for contact with the earth. (4) Micro thrusters — changing the direction and speed of the spacecraft and keeping it in the correct orbit. (5) GPS receiver — finding the place of the spacecraft.

Question: Activity: Make a model of an artificial satellite / spacecraft (with cardboard, coloured paper, scissors and glue).

Answer: Make the body of the spacecraft (space bus) from cardboard, fix solar panels (shiny blue paper) on both sides, fix a model of an antenna and a camera on top, and put a label with the name on each part.

Purpose: explain the purpose of your satellite (example: communication,

weather news, pictures of the earth). Tell whether it will be sent to the Moon or to Mars, and why. Explain the working of the parts: the solar panels give electricity, the antennas contact the earth, the cameras take pictures, and the thrusters keep the craft in the correct orbit.